MADISYN BROOKS

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madisyn-brooks.github.io

SUMMARY

Detail-oriented physics PhD student and astrophysics researcher with strong technical expertise in Python, data analysis, and statistical modeling, with additional experience in SQL, Git, Jupyter, LaTeX, and scientific computing. Proficient at communicating technical concepts through peer-reviewed publications (600+ citations), presentations, and public outreach, with a strong foundation in physics and problem solving.

EDUCATION

University of Connecticut *2023 - May 2028 (expected)*
PhD in Physics
Advisor: Jonathan Trump

University of Central Florida 2019 - 2023
Bachelors of Science in Physics
Cum Laude

RESEARCH POSITIONS

NSF Graduate Research Fellow *August 2025 - Present*
University of Connecticut

Graduate Research Assistant 2023 - 2025
University of Connecticut

Research and Mentoring Program Scholar 2021 - 2023
University of Central Florida

Research Experience for Undergraduate (REU) Summer Researcher Summer 2022
University of California-Davis

Research Experience for Undergraduate (REU) Summer Researcher Summer 2021
National Radio Astronomy Observatory (NRAO)

PUBLICATIONS, AS FIRST AUTHOR

Brooks, M. *RIPPLES in the OCEANS: Broad Line Variability of Little Red Dots*, ApJ submitted, arXiv:2608.12487

Brooks, M., *Beyond the Monsters: A Complete Census of Black Hole Activity at Cosmic Dawn*, ApJ ApJ 1002, 129 (2026)

Brooks, M., *Here There Be (Dusty) Monsters: High-redshift Active Galactic Nuclei Are Dustier than Their Hosts*, ApJ 986, 117 (2025)

PUBLICATIONS, AS SECOND AUTHOR

Davis, K., **Brooks, M.**, *OCEANS of Absorption: High-resolution NIRSpec Spectroscopy Reveals Diverse Balmer-line Absorption in Little Red Dots,* ApJ submitted, arXiv:2606.00258

Davis, K., **Brooks, M.**, *Extreme Emission Line Galaxies in CEERS Are Powered by Star Formation, not AGN*, ApJ submitted, arXiv:2602.23310

PUBLICATIONS, AS NTH AUTHOR

Roy, N. C., *Little Red Dots on FIRE: Exploring the formation and observational signatures of ultra-compact early galaxies*, ApJ submitted, arXiv:2606.23683

Gupta, A., *A Rapid Evolution in the Observed Mbh/M* Relation at $z > 3$ Revealed via Spectro-photometric SED-Modeling*, ApJ submitted, arXiV:2605.30414

Barro, G., *From "the Cliff" to "Virgil": Mapping the Spectral Diversity of Little Red Dots with JWST/NIRSpec,* ApJ 1003, 96 (2026)

Taylor, A. J., *CAPERS-LRD-z9: A Gas-enshrouded Little Red Dot Hosting a Broad-line Active Galactic Nucleus at $z = 9.288$*, ApJ 989L, 7 (2025)

Taylor, A. J., *Broad-line AGNs at $3.5 < z < 6$: The Black Hole Mass Function and a Connection with Little Red Dots*, ApJ 986, 165 (2025)

Finkelstein, S. L., *The Cosmic Evolution Early Release Science Survey (CEERS)*, ApJ 983L, 4 (2025)

Backhaus, B. E., *Emission-Line Diagnostics at $z > 4 : [OIII]\lambda 4363/H\gamma$*, ApJ submitted, arXiv:2502.03519

Davis, K., *A Census from JWST of Extreme Emission-line Galaxies Spanning the Epoch of Reionization in CEERS*, ApJ 974, 42 (2024)

AWARDED GRANTS, AS CO-I

Cycle 4 JWST, GO-8410
A Census of Galaxy Kinematics and Outflows to $z \sim 7$
PI Raymond Simons

Cycle 4 JWST, GO-8559
SPAM: Star-formation from Photometry through the Addition of Medium-bands
PI Kelcey Davis, Co-PI Rebecca Larson

SELECTED AWARDS AND HONORS

National Science Foundation Graduate Research Fellowship	*August 2025 - Present*
Connecticut Women of Innovation Finalist	Fall 2026
UCF College of Sciences General Scholarship	Spring 2023
Society of Physics Students Future Faces of Physics Award	Fall 2021, Fall 2022
Society of Physics Students Marsh White Award	Fall 2021, Fall 2022
Sigma Pi Sigma Member	2022 - 2023
UCF Pegasus Scholarship	2019 - 2023
Florida Bright Futures Scholarship	2019 - 2023

SELECTED ORAL PRESENTATIONS

Brooks, M. "An Intrinsic Black Hole Census from the JWST Extragalactic Deep", *GalFirenze 2026*, Florence, Italy.

Brooks, M. "Intrinsic BH Populations of the High Redshift Universe," *CEERS and CAPERS Collaboration Meeting 2026*, Bar Harbor, ME.

Brooks, M., "Updates from JWST Science in the Extragalactic Deep Fields", *JWST Weekly Science Briefing April 2025*, Virtual.

Brooks, M., R. Simons, J.R. Trump, "Dust Attenuation of BLAGN in the JWST Deep Fields," *University of Connecticut Astronomy Seminar Fall 2024*, Storrs, CT.

Brooks, M., R. Simons, J.R. Trump, "Dust Attenuation of BLAGN in the JWST Deep Fields," *2024 Tinsley Workshop*, New Haven, CT.

Brooks, M., D. Perlot, M. Guthrie, K. Davis,"The Restoration of the Cynthia Peterson Memorial Planetarium," *Middle Atlantic Planetarium Society (MAPS) Conference 2024*, Mystic, CT.

Brooks, M., R. Simons, J.R. Trump,"Dust Attenuation of Little Red Dots," *CEERS Collaboration Meeting 2024*, San Lorenzo de El Escorial, Spain.

SKILLS

Python	NumPy, SciPy, AstroPy, Matplotlib, Pandas, emcee, Seaborn
Machine Learning	scikit-learn, PyTorch, Clustering, Dimensionality Reduction
Tech	Jupyter Notebook (Anaconda), SQL, Git, LaTeX, VSCode, Windows
Statistics	Bayesian Inference, Statistical Analysis, Regression Models
Working Knowledge	IDL, HTML, C, Linux

TEACHING EXPERIENCE

Undergraduate Academic and Research Mentor, University of Connecticut
Jennifer Gearin (UConn BSc '28) — Spring 2026 - *Present*
Yarden Sackett (UConn BSc '25) — Fall 2024 - Spring 2025

Guest Lecturer, University of Connecticut
PHYS4730, *Cosmology and General Relativity*, Friedman Equation, — 02/10/2026
PHYS1025Q, *Introductory Astronomy*, Moon Phases, — 09/10/2024
PHYS1025Q, *Introductory Astronomy*, Seasons, — 09/05/2024

Graduate Teaching Assistant, University of Connecticut
Introductory Astronomy Laboratory, PHYS1025Q — Fall 2023, Fall 2024, Spring 2025
General Physics II Laboratory, PHYS1202Q — Spring 2024
Physics BRIDGE program — Summer 2025

OUTREACH AND SERVICE

PENGUIN Mentoring Founding Member — *Summer 2025 - Present*

Graduate Student Budget Committee, University of Connecticut — *Fall 2026 - Present*

Planetarium Presenter, University of Connecticut — *Fall 2023 - Present*

Observatory Volunteer, University of Connecticut — *Fall 2023 - Present*

Astronomy Seminar Committee, University of Connecticut — *Spring 2024 - Present*

Executive Secretary, NASA Review Panel — Summer 2025

AI Safety Group (BEACON), University of Connecticut
Social Media Coordinator — Fall 2024 - Spring 2025

Women in Physics Society, University of Central Florida
President — Fall 2021 - Spring 2023
Secretary — Spring 2021
CuWiP Organizing Committee — Fall 2021 - Spring 2023

Society of Physics Students, University of Central Florida
Treasurer — Fall 2021 - Spring 2023